

Cấu hình Prometheus

sudo vi /etc/prometheus/prometheus.yml

```
global:
  scrape_interval: 15s

scrape_configs:
- job_name: prometheus
  static_configs:
    - targets:
      - localhost:9090
- job_name: node
  static_configs:
    - targets:
      - localhost:9100
      - 192.168.232.153:9100
```

Tạo systemd service

sudo vi /etc/systemd/system/prometheus.service

```
[Unit]
Description=Prometheus
After=network-online.target

[Service]
User=prometheus
Group=prometheus

ExecStart=/usr/local/bin/prometheus \
  --config.file=/etc/prometheus/prometheus.yml \
  --storage.tsdb.path=/var/lib/prometheus \
  --web.console.templates=/etc/prometheus/consoles \
  --web.console.libraries=/etc/prometheus/console_libraries

Restart=always

[Install]
WantedBy=multi-user.target
```

Cài và tạo service node_exporter

sudo vi /etc/systemd/system/node_exporter.service

```
[Unit]
Description=Prometheus Node Exporter
After=network-online.target

[Service]
User=node_exporter
Group=node_exporter
Type=simple

ExecStart=/usr/local/bin/node_exporter

Restart=always

[Install]
WantedBy=multi-user.target
```

Reload service

sudo systemctl daemon-reload

sudo systemctl enable --now node_exporter

```
zabbix@zabbix:~$ curl -s 'http://localhost:9090/api/v1/query?query=up'
{"status":"success","data":{"resultType":"vector","result":[{"metric":{"__name__":"up","instance":"localhost:9090"},"job":"prometheus"},"value":[1786344690.126,"1"]},{metric":{"__name__":"up","instance":"localhost:9100"},"job":"node"},"value
```

Cài Grafana

sudo apt install -y apt-transport-https software-properties-common wget

sudo mkdir -p /etc/apt/keyrings/

wget -q -O - https://apt.grafana.com/gpg.key | gpg --dearmor | sudo tee
/etc/apt/keyrings/grafana.gpg > /dev/null

echo "deb [signed-by=/etc/apt/keyrings/grafana.gpg] https://apt.grafana.com stable main" | \

sudo tee -a /etc/apt/sources.list.d/grafana.list

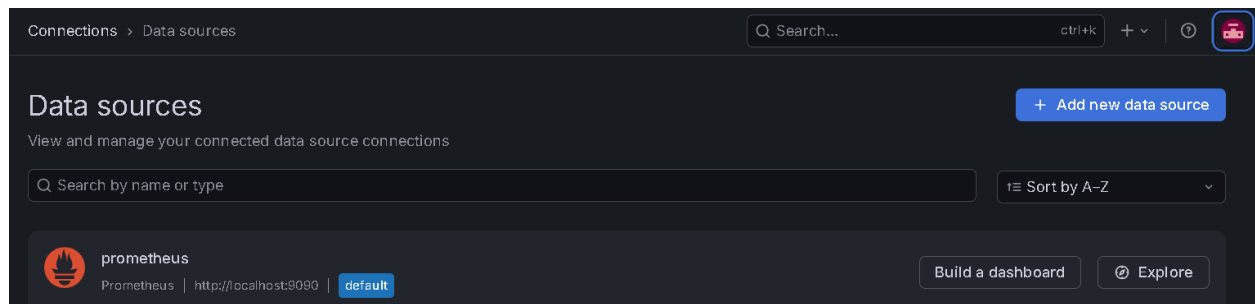
sudo apt update

```

    . prometheus }, value : [1788344090.128, 1 ]}, { metric : { __name__ : up , instance : localhost:9100 , job : node
zabbix@zabbix:~$ sudo systemctl status grafana-server
● grafana-server.service - Grafana instance
   Loaded: loaded (/usr/lib/systemd/system/grafana-server.service; enabled; preset: enabled)
   Active: active (running) since Mon 2026-08-10 03:06:04 UTC; 3h 50min ago
     Docs: http://docs.grafana.org
   Main PID: 1901 (grafana)
    Tasks: 32 (limit: 2205)
   Memory: 236.3M (peak: 465.3M)
      CPU: 2min 15.251s
   CGroup: /system.slice/grafana-server.service
           └─1901 /usr/share/grafana/bin/grafana server --config=/etc/grafana/grafana.ini --pidfile=/run/graf
              1909 /var/lib/grafana/plugins/elasticsearch/gpx_grafana_elasticsearch_datasource_linux_amd64
              1916 /var/lib/grafana/plugins/zipkin/gpx_grafana_zipkin_datasource_linux_amd64

```

Đăng nhập vào Grafana

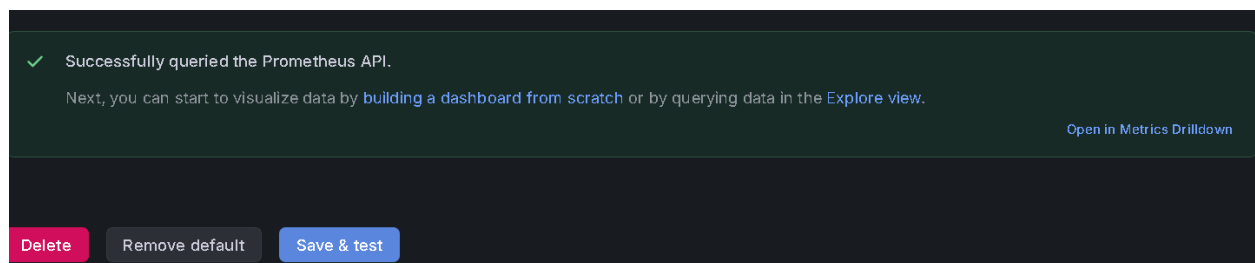


Tạo Kết Nối Grafana với Prometheus

Connections → Data sources ➔ add new data source → Prometheus

Add local host <http://localhost:9090>

Save và test kết nối thành công

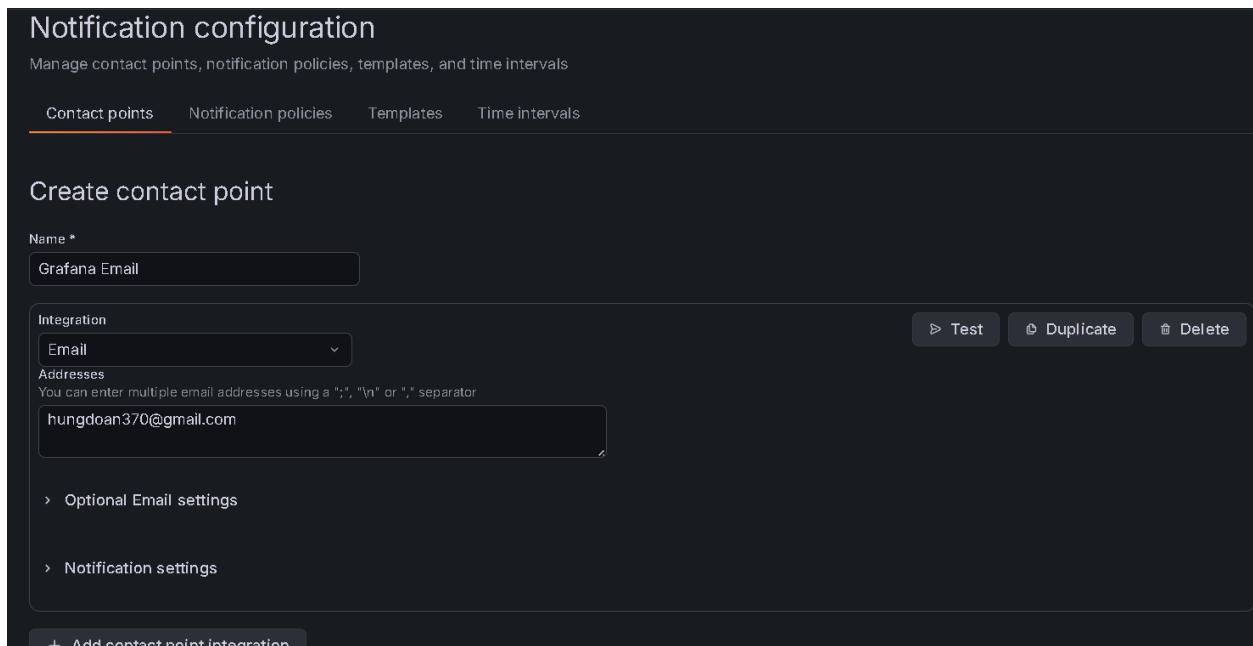


Cấu hình alert qua mail trên Grafana

Vào: **Alerting** → **Notification configuration** → **Contact points**

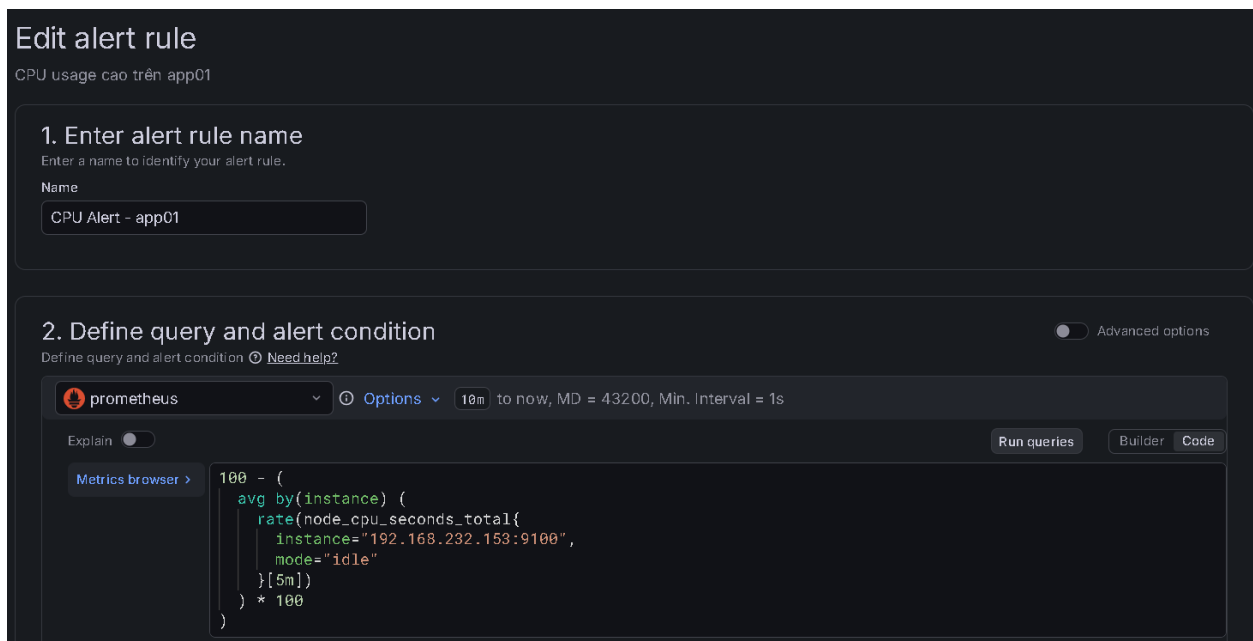
Sau đó: **Add contact point**

→ cấu hình thông tin email gửi



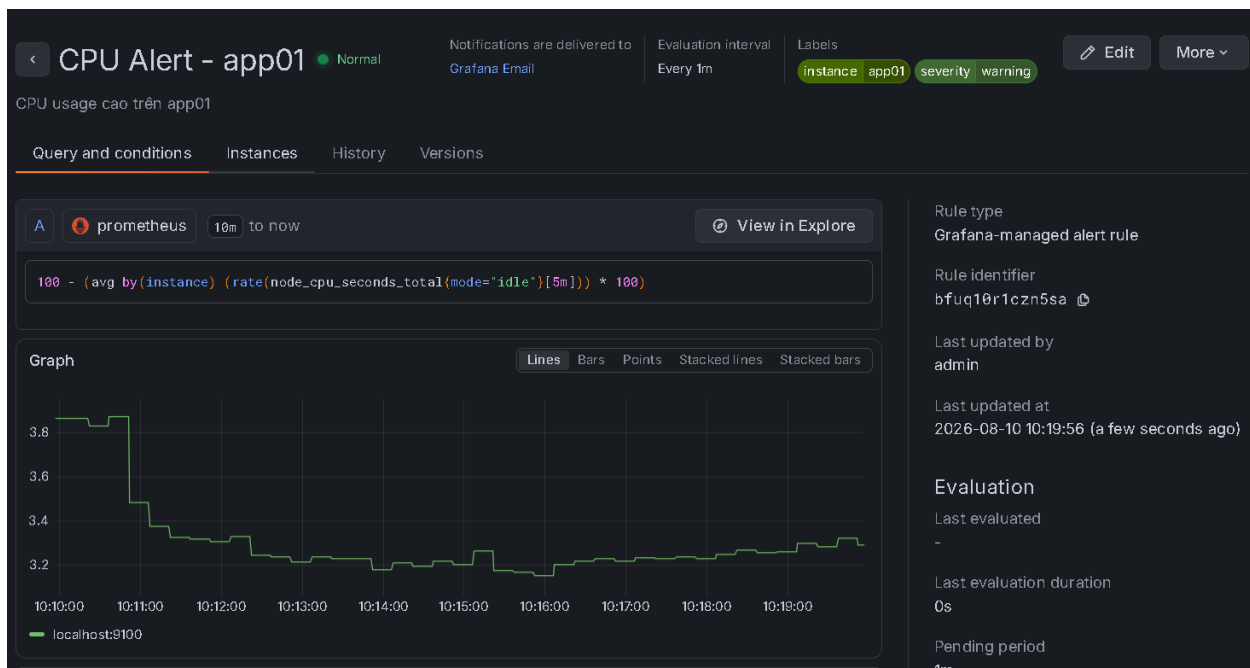
The screenshot shows the 'Notification configuration' page in Grafana. The page has a dark theme and a header with the title 'Notification configuration' and a subtitle 'Manage contact points, notification policies, templates, and time intervals'. Below the header are four tabs: 'Contact points', 'Notification policies', 'Templates', and 'Time intervals'. The 'Contact points' tab is selected. The main content area is titled 'Create contact point'. It contains a form with the following fields: 'Name' (required, with a red asterisk) containing 'Grafana Email'; 'Integration' (a dropdown menu set to 'Email'); 'Addresses' (a text area containing 'hungdoan370@gmail.com' with a note 'You can enter multiple email addresses using a ";", "\n" or "," separator'); 'Optional Email settings' (a collapsed section); and 'Notification settings' (a collapsed section). On the right side of the form are three buttons: 'Test', 'Duplicate', and 'Delete'. At the bottom left is a button '+ Add contact point integration'.

Vào: **Alerting** → **Alert rules** : Chọn: **New alert rule**



The screenshot shows the 'Edit alert rule' page in Grafana. The page has a dark theme and a header with the title 'Edit alert rule' and a subtitle 'CPU usage cao trên app01'. Below the header are two main sections: '1. Enter alert rule name' and '2. Define query and alert condition'. The '1. Enter alert rule name' section has a subtitle 'Enter a name to identify your alert rule.' and a 'Name' field containing 'CPU Alert - app01'. The '2. Define query and alert condition' section has a subtitle 'Define query and alert condition' and a 'Need help?' link. It contains a 'prometheus' dropdown menu, an 'Options' dropdown menu, and a '10m' time range selector. Below these are 'Run queries', 'Builder', and 'Code' buttons. The 'Code' button is selected, showing a Prometheus query in a text area:

```
100 - (
  avg by(instance) (
    rate(node_cpu_seconds_total{
      instance="192.168.232.153:9100",
      mode="idle"
    })[5m])
  ) * 100
```



Test cpu data overload

```
cluster4@app01:~$ stress-ng --cpu 2 --timeout 180s
stress-ng: info: [6732] setting to a 3 mins, 0 secs run per stressor
stress-ng: info: [6732] dispatching hogs: 2 cpu
```

Gửi mail về khi cpu > 80%

Problem: CPU usage is too high on app01 [Hộp thư đến x](#)



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đến tôi ▾



Có vẻ như thư này được viết bằng Tiếng Anh

[Dịch sang Tiếng Việt](#)

Problem started at 03:42:28 on 2026.08.10

Problem name: CPU usage is too high on app01

Host: dev

Severity: High

Operational data: 100 %

Original problem ID: 171

Mail báo khi cpu đã được xử lý



Zabbix-Grafana-Lab › DatasourceNoData

✓ 1 resolved instances

Resolved

DatasourceNoData

[View alert](#)

Summary

CPU usage cao trên app01

Description

CPU usage trên app01 đã vượt 80%.
Kiểm tra tiến trình đang sử dụng CPU cao.

